

AMENDMENT NO. 3 JANUARY 2025
TO
IS 3024 : 2015 COLD ROLLED GRAIN ORIENTED ELECTRICAL STEEL
STRIP AND SHEET DELIVERED IN FULLY PROCESSED STATE
(Third Revision)

(Page 2, clause 5.1) — Substitute the following for the existing:

‘5.1 The grades covered by this standard are classified according to the value of maximum specific total loss in watts per kilogram and according to the nominal thickness of the material (0.18 mm, 0.20 mm, 0.23 mm, 0.27 mm, 0.30 mm and 0.35 mm).’

(Page 4, Table 1) — Substitute the following for the existing:

‘Table 1 Magnetic Properties of Conventional Grain Oriented (CGO) Electrical Steel

(Clauses 4.2, 5.2, 8.1 and 8.2)

Sl No.	Grade	Nominal Thickness mm	Maximum Specific Core Loss at 1.5 T W/kg		Maximum Specific Total Loss at 1.7 T W/kg		Minimum Polarization in Tesla at a Field Strength of 800 A/m	Minimum Stacking Factor
			50 Hz	60 Hz	50 Hz	60 Hz		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	18CG100	0.18	0.68	0.90	1.00	1.35	1.84	0.930
ii)	23CG110	0.23	0.73	0.96	1.10	1.45	1.78	0.945
iii)	23CG120	0.23	0.77	1.01	1.20	1.57	1.78	0.945
iv)	23CG127	0.23	0.80	1.03	1.27	1.65	1.75	0.945
v)	27CG120	0.27	0.80	1.07	1.20	1.58	1.78	0.950
vi)	27CG130	0.27	0.85	1.12	1.30	1.68	1.78	0.950
vii)	27CG140	0.27	0.89	1.15	1.40	1.75	1.75	0.950
viii)	30CG120	0.30	0.83	1.09	1.20	1.58	1.78	0.955
ix)	30CG130	0.30	0.85	1.15	1.30	1.71	1.78	0.955
x)	30CG140	0.30	0.92	1.21	1.40	1.83	1.78	0.955
xi)	30CG150	0.30	0.97	1.25	1.50	1.90	1.75	0.955
xii)	35CG145	0.35	1.03	1.36	1.45	1.91	1.78	0.960
xiii)	35CG155	0.35	1.07	1.41	1.55	2.04	1.78	0.960
xiv)	35CG165	0.35	1.11	1.52	1.65	2.12	1.75	0.960

NOTE — Normally the tests should be performed at 50 Hz. However, countries where power supply at 50 Hz is not available, testing may be carried out at 60 Hz and accordingly product shall conform to the specified values given in above table. However, in all such cases, the product shall also conform to the specified values of table when tested in importing country having power supply at 50 Hz.

Amendment No. 3 to IS 3024 : 2015

[Page 4, Table 2, (see also Amendment No. 1 and Amendment No. 2)] — Substitute the following for the existing:

‘Table 2 Magnetic Properties of High Permeability Grain Oriented (HPGO) Electrical Steel
(Clauses 4.2, 5.2, 8.1 and 8.2)

Sl No.	Grade	Nominal Thickness, mm	Maximum Specific total loss at 1.7 T W/kg		Minimum Polarization in Tesla at a Field Strength of 800 A/m	Minimum Stacking Factor
			50 Hz	60 Hz		
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	20HP70 ^d	0.20	0.70	0.92	1.85	0.940
ii)	20HP75 ^d	0.20	0.75	0.99	1.85	0.940
iii)	23HP75 ^d	0.23	0.75	0.99	1.85	0.945
iv)	23HP80 ^d	0.23	0.80	1.04	1.85	0.945
v)	23HP85 ^d	0.23	0.85	1.12	1.85	0.945
vi)	23HP90 ^d	0.23	0.90	1.19	1.85	0.945
vii)	23HP95 ^d	0.23	0.95	1.25	1.85	0.945
viii)	23HP100 ^d	0.23	1.00	1.32	1.85	0.945
ix)	27HP85 ^d	0.27	0.85	1.12	1.85	0.950
x)	27HP90 ^d	0.27	0.90	1.19	1.85	0.950
xi)	27HP95 ^d	0.27	0.95	1.25	1.85	0.950
xii)	27HP100	0.27	1.00	1.32	1.88	0.950
xiii)	27HP110	0.27	1.10	1.45	1.88	0.950
xiv)	30HP95	0.30	0.95	1.25	1.88	0.955
xv)	30HP100	0.30	1.00	1.32	1.88	0.955
xvi)	30HP105	0.30	1.05	1.38	1.88	0.955
xvii)	30HP110	0.30	1.10	1.46	1.88	0.955
xviii)	30HP120	0.30	1.20	1.58	1.88	0.955
xix)	35HP110	0.35	1.10	1.45	1.88	0.960
xx)	35HP115	0.35	1.15	1.51	1.88	0.960
xxi)	35HP125	0.35	1.25	1.64	1.88	0.960
xxii)	35HP135	0.35	1.35	1.77	1.88	0.960

NOTES

1 Normally the tests should be performed at 50 Hz. However, countries where power supply at 50 Hz is not available, testing may be carried out at 60 Hz and accordingly product shall conform to the specified values given in above table. However, in all such cases, the product shall also conform to the specified values of Table when tested in importing country having power supply at 50 Hz.

2 Grades may be delivered in domain or non-domain refined condition. In case if it is delivered with domain refinement then it is necessary to apply suffix ‘d’. Suffix ‘d’ indicates use of test method to certify a grade, in line with test methods as defined in **15.4**.

3 In cases where material gets deteriorated when subjected to heat treatment, the domain refined grades need to be checked by Single sheet method as given in **14.5** and as per IS 649. For other grades/other types of domain refined grades which do not deteriorate when subjected to heat treatment, the test method shall be as Epstein as given in **14.1** and as per IS 649.’

Amendment No. 3 to IS 3024 : 2015

(Page 5, clause 10.1) — Substitute the following for the existing:

‘10.1 Stacking Factor

The stacking or lamination or space factor shall be as high as practicable consistent with the surface type and material thickness desired. The typical stacking factor values for the various material thickness and surface types, as determined by quality control tests made in accordance with test method described in IS 649, are shown in Table 4. However, in this case, the test specimen shall be of minimum 280 mm length and width of 30 mm; and shall conform to the specified dimensions within ± 0.8 mm. For thicknesses not covered in IS 649, the test specimen shall consist of at least 28 strips of the same size. The test strips shall be carefully deburred before the test. Epstein strips may be used for this test.

Table 4 Minimum Stacking Factor

(Clause 10.1)

Sl No.	Condition	Surface Type	Stacking Factor, Percent at Nominal Thickness					
			0.18	0.20	0.23	0.27	0.30	0.35
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	NF	C-2	94.0	95.0	95.5	96.0	96.0	96.5
ii)	F	C-5 Over C-2	93.0	94.0	94.5	95.0	95.5	96.0
iii)	PQ	C-5 Only	94.0	95.0	95.5	96.0	96.0	96.5'

(Page 5, clause 11.1) — Substitute the following for the existing:

‘11.1 Thickness Tolerance

The nominal thicknesses of the product are 0.18 mm, 0.20 mm, 0.23 mm, 0.27 mm, 0.30 mm, and 0.35 mm. For thickness tolerance, a distinction is made between:

- The deviation from the nominal thickness within the same acceptance unit;
- The difference in thickness in a sheet or in a length of strip in a direction parallel to the direction of rolling.
- The difference in thickness in a direction perpendicular to the direction of rolling. This tolerance applies only to products with a width greater than 150 mm;
- At any point, the deviation from the nominal thickness within the same acceptance unit shall not exceed the tolerances of Table 6. The additional thickness due to welds with respect to the measured thickness of the product shall not exceed 0.050 mm;
- The difference in thickness in a sheet or in a length of strip of 1 m in a direction parallel to the direction of rolling shall not exceed 0.030 mm; and
- For products with a width greater than 150 mm, the difference in thickness in a direction perpendicular to the direction of rolling shall not exceed 0.020 mm, the measurements being made at least 20 mm from the edges. For narrow strips, other agreements may be needed.'

(Page 6, Table 5) — Insert the following new table after Table 5:

‘Table 6 Tolerances on Nominal Thickness

(Clause 11.1)

Sl No.	Nominal Thickness mm	Tolerance mm
(1)	(2)	(3)
i)	0.18	± 0.020
ii)	0.20	± 0.020

Amendment No. 3 to IS 3024 : 2015

Table 6 (Concluded)

Sl No.	Nominal Thickness mm	Tolerance mm
(1)	(2)	(3)
iii)	0.23	± 0.023
iv)	0.27	± 0.027
v)	0.30	± 0.030
vi)	0.35	± 0.030

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